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**B.Sc. (E.C.S.) (Semester - II) (NEP CBCS) Examination:
March/April - 2026
Discrete Mathematics (G06-0204)**

Day & Date: Saturday, 11-04-2026
Time: 03:00 PM To 04:30 PM

Max. Marks: 30

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.
3) Use of scientific calculator is allowed.

Q.1 Choose the Correct Alternative.

06

- 1) If $|A| = 35$, $|B| = 20$ and $|A \cap B| = 15$, Then $|A \cup B| = \underline{\hspace{2cm}}$.

a) 84 b) 170
c) 40 d) 20
- 2) If $A = \{a, b, c, e, f, h\}$ then $|A| = \underline{\hspace{2cm}}$.

a) 4 b) 5
c) 6 d) 7
- 3) The function which is injective & surjective is _____ function.

a) Bijective b) Surjective
c) Identity d) Injective
- 4) If $f(x) = 2x^2 - x + 5$ then $f(-2)$ is _____.

a) 15 b) 11
c) -1 d) 0
- 5) Let R be the relation defined on the set $A = \{a, b, c, d\}$ given by $R = \{(a, b), (b, a), (a, d), (b, b), (b, d), (c, a), (c, d), (d, b), (d, a)\}$. Order of matrix of relation R is _____.

a) 3×3 b) 2×2
c) 4×4 d) 5×5
- 6) Find the value of a_4 for the recurrence relation $a_n = 2a_{n-1} + 3$, with $a_0 = 6$.

a) 320 b) 221
c) 141 d) 65

Q.2 Attempt the following. (Any Three)

06

- Find the order of recurrence relation $a_r - 7a_{r-1} + 10a_{r-2} = 0$
- Define finite set with example.
- State the pigeonhole principle.
- Define equivalence relation.
- Define domain of function.

Q.3 Attempt the following. (Any Two) 06

- a) Let $f: R \rightarrow R$ is defined by $f(x) = \frac{5x-4}{3}$ show that $f(x)$ is bijective function.
- b) Define reflexive relation & symmetric relation.
- c) Let R is a relation defined on set $A = \{1,2,3\}$ & $R = \{(1,1), (1,2), (2,3), (3,1), (3,2)\}$ find transitive closure of R by Wharshall's algorithm.

Q.4 Attempt the following. (Any Two) 06

- a) Define homogeneous recurrence relation with constant coefficient & find characteristic equation of recurrence relation

$$a_r - 8a_{r-1} + 16a_{r-2} = 0$$
- b) What is injective & surjective function?
- c) If $f(x) = x^2 + x - 1$ then find
 i) $f(2)$, ii) $f(-4)$, iii) $f(2x)$

Q.5 Attempt the following. (Any One) 06

- a) State & prove mutual inclusion-exclusion principle for 3 sets.
- b) Solve the recurrence relation $a_r - 5a_{r-1} + 6a_{r-2} = 0$ with initial conditions $a_0 = a_1 = 1$.